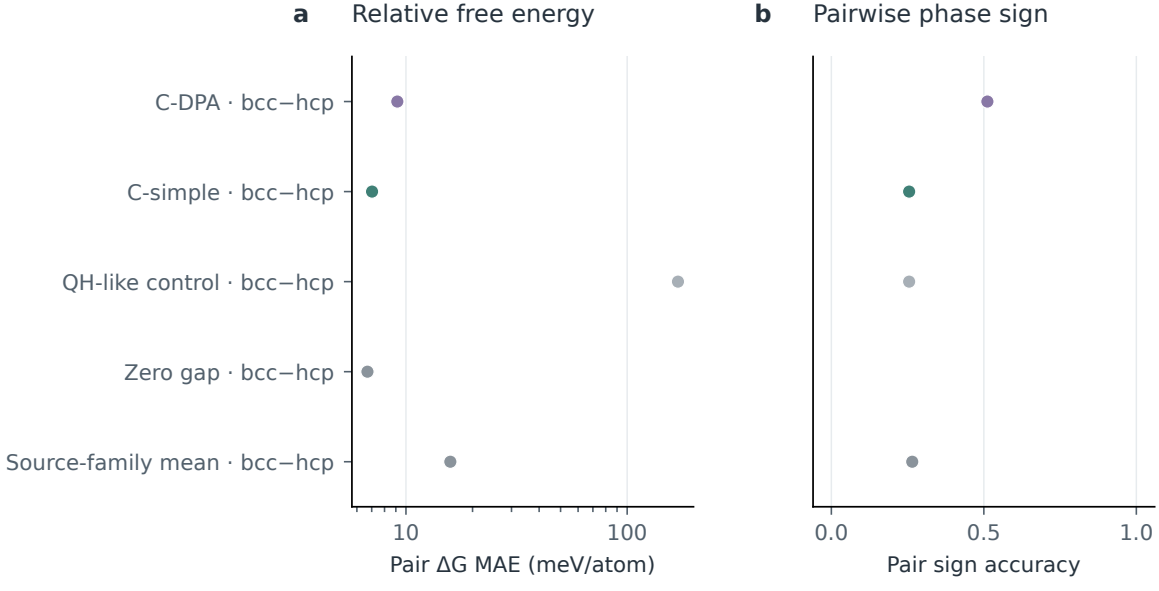
a
Relative free energy
C-DPA · bcc−hcp
C-simple · bcc−hcp
QH-like control · bcc−hcp
Zero gap · bcc−hcp
Source-family mean · bcc−hcp
10
100
Pair ΔG MAE (meV/atom)
b
Pairwise phase sign
0.0
0.5
1.0
Pair sign accuracy